

EX:NO:28

PROGRAM:

```
import itertools
```

```
def knapsack(weights, values, capacity):  
    n = len(weights)  
    max_value = 0  
    best = None  
  
    for r in range(n+1):  
        for comb in itertools.combinations(range(n), r):  
            total_w = sum(weights[i] for i in comb)  
            total_v = sum(values[i] for i in comb)  
  
            if total_w <= capacity and total_v > max_value:  
                max_value = total_v  
                best = comb  
  
    return best, max_value
```

Test

```
print(knapsack([2,3,1],[4,5,3],4)) # ([0,2],7)
```

```
print(knapsack([1,2,3,4],[2,4,6,3],6)) # ([0,1,2],10)
```

Output

```
((1, 2), 8)
```

```
((0, 1, 2), 12)
```

```
=== Code Execution Successful ===
```